

Introduction to Discrete Mathematics

Probability and Counting

- The probability of an event E is the number of outcomes in E , $N(E)$ divided by the total number of possible outcomes, $N(S)$.
- **Multiplication Rule:** If you have a process that can be completed in k steps where each step can be done in $n_i, i = 1, \dots, k$, ways, then the process can be completed in $n_1 \times n_2 \times \dots \times n_k$ ways.
- **Permutation:** A *permutation* of a set S is an ordering of the elements of S . For example, 3, 1, 2 (often written 312) is a permutation of $S = \{1, 2, 3\}$.
- **r -Permutation:** An *r -permutation of a set of n elements* is an ordered set of r elements taken from a set of n elements. For example, 421 is a 3-permutation of $S = \{1, 2, 3, 4, 5\}$.
- The number of r -permutations of a set of n elements is denoted $P(n, r)$.

1. Suppose you toss a fair coin two times.

- (a) List all the possible outcomes.
- (b) What is the probability that you get 2 heads?
- (c) What is the probability that you get 1 head?

2. Suppose a restaurant offers 3 appetizers, 3 entrees, and 2 desserts.

- (a) Draw a possibility tree for how many different meals (one appetizer, one entree, and one dessert) you can create.

- (b) How many total meals can you create?

3. Suppose a license plate consists of 3 letters followed by 2 digits.

(a) How many license plates can you make?

(b) How many license plates can you make if you cannot repeat a letter or number?

4. List all the permutations of $S = \{1, 2, 3\}$. Hint: there are 6.

5. How many permutations are there of a set with 5 elements?

6. How many permutations are there of a set with n elements?

7. How many 2-permutations are there of $S = \{1, 2, 3\}$?

8. How many 3-permutations are there of $S = \{1, 2, 3, 4, 5\}$? Think of finding a way to count them using the multiplication rule rather than writing them all down.

9. Try to generalize to find a formula for counting all r permutations in a set of n elements.

10. Find $P(10, 8)$.